Uroš Pantelić

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Electrical Engineering graduate (University of Belgrade, August 2026) with industry experience spanning backend systems, ML infrastructure, and large-scale distributed platforms. Interned at **Microsoft** and **Tenstorrent** working on LLM inference and training systems; built production backend services and full-stack applications independently. Strong foundation in algorithms, data structures, and systems programming. Passionate about building reliable, scalable software and solving hard engineering problems end to end.

EXPERIENCE

Nextesy May 2026 – Aug 2026

Software Engineer Intern, Backend

- Designing and implementing a **role-based permission system** for a multi-tenant backend, supporting fine-grained access control across user roles, resources, and organizational scopes.
- Optimizing permission resolution through **indexed DB queries and query batching** to minimize authorization overhead on high-traffic endpoints.
- Building an **integration and load testing suite** that verifies access-rule correctness and benchmarks endpoint throughput under concurrent load, catching permission regressions before they reach production.

Tenstorrent Feb 2025 – Nov 2025

Machine Learning Intern

- Enabled **parameter-efficient fine-tuning** (LoRA/DoRA) for LLaMA 3.2 1B by extending training pipelines for scalable multi-device adaptation.
- Built reproducible training and evaluation pipelines; benchmarked latency, throughput, tokens/sec, and memory utilization across device configurations.

Open Source Contributor, TT-XLA

- Implemented **tensor-parallel inference** for LLaMA 3.1 8B in JAX/Flax, compiled through StableHLO into Tenstorrent's **MLIR-based compiler stack** (TT-XLA); validated numerical parity against PyTorch/HuggingFace baselines and contributed the implementation upstream.

Microsoft Aug 2024 – Nov 2024

Machine Learning Engineer Intern

- Researched selective state-space models (Mamba) and built internal prototypes in PyTorch; constructed a benchmarking harness against transformer baselines (LLaMA/GPT-style).
- Benchmarked latency, throughput, memory usage, and quality metrics; delivered a technical report that informed model selection and deployment trade-offs.

PROJECTS

Cloud-Native Distributed Inference Platform Python, FastAPI, WebSockets, Docker, Redis

- Designed and built a distributed job orchestration system for CNN inference, modeled on cloud-native scheduler architecture (Kubernetes-style job queues with pluggable executors).
- Implemented worker pool management with **dynamic scaling, DAG-based dependency resolution, task prioritization**, and automatic fault recovery with configurable retry logic.
- Built a real-time observability dashboard (WebSockets) exposing per-node load, task throughput, queue depth, and retry rates; benchmarked latency and throughput under varying concurrency, achieving near-linear scaling.

AI Agent Orchestrator (Personal Project, 2026) Python, Claude Code, tmux, Git

- Built a Python CLI and daemon that runs multiple LLM coding agents in parallel, each in its own git worktree and tmux session, pulling tasks from a shared queue with per-task locking.
- Implemented the full task lifecycle (dispatch, park, resume, cancel): parking commits work in progress and releases the lock, so a task can later resume on any idle worker without losing state.
- Added deterministic recovery from outages and self-healing of leaked locks, plus a web UI board and per-worker logs for monitoring running agents.

Study Feed, AI Study App (Personal Project, 2026) Flutter, Dart, Riverpod, Drift, Claude API

- Built a mobile app that turns course materials (PDFs, text, images, YouTube links) into summaries, quizzes, and short educational character dialogs, studied through a vertical TikTok-style feed.
- Designed the content generation pipeline around the Claude API and ElevenLabs text-to-speech, with word-synced captions timed from the TTS output and cached background video loops.
- Made it local-first with a Drift (SQLite) database and a spaced-repetition review ladder with streaks; optional Google sign-in backs up progress to Firebase.

COMPETITIONS & HACKATHONS

Qualcomm LPCVC 2025, Track 1 3rd Place

- Image classification under real-world distribution shifts; compiled and profiled models on Qualcomm AI Hub (Snapdragon 8 Elite QRD) for on-device inference. Invited to present at **CVPR 2025** (Nashville).

DigiHack Hackathon 2nd Place

Developed a predictive maintenance solution for industrial equipment using sensor data and anomaly detection.

ASEF Hackathon 2nd Place

Built an air pollution forecasting system for Belgrade using weather-based features and regression models.

EDUCATION

University of Belgrade 2021 – Aug 2026

B.Sc. Electrical Engineering, Signals and Systems, with Computer Science electives **GPA: 9.0 / 10.0**

Relevant coursework: Algorithms & Data Structures, Operating Systems, Computer Networks, Machine Learning, Computer Vision, Electronics, Data & Signal Processing, Linear Algebra, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript/TypeScript, Dart, SQL
Backend & Systems: Django, FastAPI, REST APIs, WebSockets, PostgreSQL, Redis, Docker, CI/CD, Git
CS Fundamentals: Algorithms & Data Structures, OOP, Concurrency, Operating Systems, Computer Networks
Distributed Systems: Task Scheduling, Load Balancing, Fault Tolerance, Sharding, System Design
ML & LLMs: PyTorch, JAX/Flax, HuggingFace, LLaMA, LoRA/DoRA, RAG, FAISS, LLM Agents, ML compilers (XLA, StableHLO, MLIR)
Mobile: Flutter, Riverpod, Drift, Firebase

CERTIFICATIONS & BOOTCAMPS

Google Machine Learning Bootcamp
Harvard CS50's Introduction to Databases with SQL
Machine Learning Specialization (Coursera)
Deep Learning Specialization (Coursera)
TryHackMe Cybersecurity Foundations

LANGUAGES

Serbian (Native) English (C1, Professional) Turkish (Basic)